Dávid Straka

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EDUCATION

MSc in Stochastics and Financial Mathematics, track Data Analysis and Statistics Amsterdam, Netherlands
Universiteit van Amsterdam *2023 – 2026*

- Relevant courses: Measure Theoretic Probability, Computational Finance, Stochastic Simulation, Asymptotic Statistics
- Thesis: Optimization with Quantum Annealing

Study abroad in MSc Fundamental Principles of Data Science, and MSc Statistics and Operations Research Barcelona, Spain
Universitat de Barcelona and Universitat Politècnica de Catalunya *2024*

- Relevant courses: Optimization, Numerical Linear Algebra, Machine Learning, Generalized Linear Models

BSc in Mathematics, track Data Science Amsterdam, Netherlands
Vrije Universiteit Amsterdam *2019 – 2023*

- GPA: 8.0/10
- Relevant courses: Statistical Data Analysis, Data Structures and Algorithms, Introduction to Programming (Java), Markov Chains
- Thesis: compared Competing risk forests ML model with a statistical Cox model in survival analysis

Bilingual Diploma of the International Baccalaureate (IB), and Slovak Secondary School Diploma Bratislava, Slovakia
Gymnázium Jura Hronca *2018*

- Higher level courses: English, Mathematics, Biology; Standard level courses: Slovak language, Chemistry, Economics

WORK EXPERIENCE

Research Project Intern The Hague, Netherlands
TNO – Netherlands Organization for Applied Scientific Research *2025*

- Reviewed ~ 10 financial risk measures and identified expected shortfall (ES) as suitable for portfolio optimization using a D-Wave quantum annealer
- Reformulated the mean-ES portfolio optimization into 2 suitable (QUBO) optimization forms and derived 6 mathematical bounds on 3 decision variables to ensure compatibility with quantum annealers
- Optimized the mean-ES portfolio of S&P 500 constituents on the D-Wave annealer and, hence, verified the suitability of using ES in portfolio annealing

Machine Learning Intern Amsterdam, Netherlands
Dutch Summer of AI at Company.info *summer 2022*

- Trained a conditional random field model within an international team of 5 interns to recognize relevant named entities in legal texts and to link Dutch companies to relevant lawsuits
- Competed in a Google hackathon at ABN AMRO; attended 2 Google sessions on AI challenges and ethical dilemmas
- Joined showcases featuring AI projects from 7 companies (ABN AMRO, AEGON, DSM, NS, Shell, SouthPole, and Tata Steel)

Calculus and Linear Algebra Teacher Assistant Amsterdam, Netherlands
Vrije Universiteit Amsterdam *2020 – 2021*

- Conducted weekly tutorials across a full academic year for ~ 20 BSc Calculus and Linear Algebra students, led solution walkthroughs, and weekly selected top 10 effective exercises for understanding
- Guided ~ 20 students in-person and via Zoom; offered individualized after-hours help by email, clarified concepts, and discussed problem-solving approaches
- Proctored online exams and co-graded ~ 100 exams bimonthly in Canvas, managed regrade requests, and corrected grading errors to ensure fair outcomes

Housing Services Assistant Amsterdam, Netherlands
DUWO *summer 2020*

- Welcomed and onboarded tens of new tenants during move-in, issued keys, explained basic procedures, and guided the tenants to the assigned accommodation

Shelf Stocker Amsterdam, Netherlands

Albert Heijn *2019 - 2022*

- Unloaded delivery trucks, sorted and stored inventory, stocked shelves, removed expired items, cleaned floors using a machine, and assisted customers in locating products

Food Delivery Rider Amsterdam, Netherlands

Uber Eats (freelancing), Gorillas *2019, 2021*

- Fulfilled restaurant and grocery orders by bicycle across Amsterdam while optimizing routes in often adverse weather conditions; navigated peak-hour demand; and ensured accurate, on-time delivery
- Operated Uber Eats work as a freelancer; administered invoicing; filed quarterly VAT returns and yearly income tax returns; communicated with customers to resolve order issues
- Registered the freelancing in the Business Register at the Chamber of Commerce; applied for and complied with the Small Businesses Scheme

English Second Language Teacher Nová Dubnica, Slovakia

BS SCHOOL language school (incl. freelancing) *2018 – 2019*

- Taught English as a second language to two adult groups of 5–15 learners in classrooms and at a corporate client, and fostered friendly, interactive, confidence-building lessons
- Facilitated high-intensity dialogue-based Direct Method lessons, adapted pacing to mixed levels, and conducted dictations; steered group dynamics and sustained an engaging, supportive learning environment
- Operated as a registered freelancer, created and administrated invoices monthly, documented lesson records and progress, marked writing homework, and graded exams with corrective feedback

Coworking Community Host Bratislava, Slovakia

Progressbar Hackerspace *summer 2018*

- Hosted guests in a coworking and hackerspace environment, welcomed visitors, offered basic orientation and support, and kept day-to-day operations smoothly

Project Coordinator Bratislava, Slovakia

Bratislava Region Youth Council *summer 2018*

- Contacted and persuaded tens of high school representatives across Slovakia to join simulated student elections, tracked responses, scheduled follow-ups, and aligned participation with project timelines and requirements
- Contacted ~ 10 prospective funders and applied for relevant grants to support a participatory civic app, prepared application materials, and tracked submissions against project priorities and deadlines
- Created and presented twice an investor-ready pitch presentation for the participatory civic app designed to streamline citizen engagement with public institutions

Logistics and Accounts Payable Assistant Trenčín, Slovakia

Akebono Brake *summer 2016*

- Updated an Excel-based logistics database of materials for brake production, and ensured accurate identification of production-line materials by creating and labelling the materials across logistics areas
- Administered invoices by recording, archiving, and entering incoming documents into the company system

Warehouse Operative Trenčín, Slovakia

Profit MK *summer 2015*

- Assembled products and unloaded incoming deliveries, supported warehouse operations, and kept the work area organized and efficient

Restaurant Crew Member Trenčín, Slovakia

McDonald's *2014 – 2015*

- Prepared menu items in a fast-paced environment, followed food-safety and hygiene procedures, operated the cash register, processed orders and payment, and provided efficient customer service during peak periods
- Cleaned and sanitized kitchen and restaurant areas, and supported smooth shift operations through reliable teamwork and task prioritization

SKILLS AND INTERESTS

- **Programming languages**: Python (sklearn, TensorFlow, CVXPY, SciPy, NumPy), R, SQL, Java, MATLAB, Mathematica
- **Languages**: Slovak (native), English (fluent), Czech (fluent comprehension), Dutch (INTT B2 Certificate)
- **Interests**: padel, piano, gym, running, cinema club, storytelling
- **Other**: driver's license

CERTIFICATES

HTML, CSS, JavaScript + PHP – programming websites Certificate Bratislava, Slovakia
IT LEARNING *2014*